PROJECT QUOTATION

AI-Powered YouTube Title Optimizer

Multi-Agent System + Machine Learning Module + Full Thesis

BSc (Hons) Software Engineering Research Project

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Institution:	NSBM Green University
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Confidential — For intended recipient only

1. Project Overview

Development of an AI-powered, multi-agent system that automatically analyzes a YouTube video's content and generates psychologically optimized, virality-engineered titles — for a BSc (Hons) Software Engineering research project at NSBM Green University.

The system covers a LangChain multi-agent pipeline (Analyzer, Generator, Critic agents), a Predicted CTR (pCTR) machine learning model (Random Forest / XGBoost), an NLP evaluation layer (VADER sentiment, BERTScore, NLI faithfulness check), and a FastAPI REST backend. Built with AI-assisted tooling and scoped as a fully functional research prototype sufficient to complete and defend the thesis.

A complete thesis document covering all chapters (Introduction through Conclusion) and 8 included mentorship sessions are provided alongside the system.

2. Deliverables

Deliverable 1 — System Prototype

Component	Details
Analyzer Agent	**Extracts core video promise, key entities, emotional potential, and applicable psychological triggers from transcript / description**
Generator Agent	**Produces 5 candidate titles per run, each targeting a different psychological trigger (curiosity gap, loss aversion, social proof, urgency, extreme framing)**
Critic Agent	**Scores candidates with pCTR model, VADER arousal check, BERTScore relevance, and NLI deception detection — auto-rejects contradictory titles**
pCTR ML Model	**Random Forest / XGBoost trained on YouTube trending dataset with engineered title features**
NLP Layer	**VADER sentiment scoring, BERTScore relevance filter, NLI faithfulness gate (prevents AI-hallucinated titles)**
FastAPI Backend	**Async REST API exposing the full pipeline as a clean endpoint**
YouTube API Integration	**Dataset collection pipeline via YouTube Data API v3**
Demo & Evaluation	**Working demo with ranked title outputs, evaluation metrics, and sample results**
Source Code	**Complete, documented — delivered via private GitHub repository**

Deliverable 2 — Full Thesis Document

Chapter	Content
Chapter 1 — Introduction	**Background, problem statement, research objectives, scope, and significance**
Chapter 2 — Literature Review	**Information Gap Theory, psychological triggers, YouTube algorithm research, clickbait studies, existing NLP tools — all cited from academic sources**
Chapter 3 — Methodology	**Research design, dataset selection and justification, system architecture rationale, agent design decisions**
Chapter 4 — System Design & Implementation	**Component diagrams, agent pipeline walkthrough, technology stack justification, ML model design**
Chapter 5 — Results & Evaluation	**pCTR model metrics, VADER scores, NLI validation results, sample title output comparison**
Chapter 6 — Conclusion	**Summary of findings, research contributions, limitations, and future work**
References & Bibliography	**IEEE format, fully cited — 20+ academic sources**

Deliverable 3 — Supporting Documentation

Item	Details
Architecture Diagrams	**Agent flow diagram, system component diagram, data flow diagram**
Evaluation Report	**Model accuracy metrics, confusion matrices, sample title outputs with scores**
README	**Setup and run instructions for the full prototype**

3. Mentorship Sessions

Included Sessions (8 sessions — provided after system delivery)

Sessions are conducted online via Google Meet / Zoom. Duration: 1–1.5 hours each. Scheduled bi-weekly or as agreed.

#	Topic	What You Will Be Able to Do After	Duration
1	Project Overview & Research Context	Explain the problem, research gap, and why this project is novel	**1 hr**
2	Theoretical Foundations	Articulate Information Gap Theory, Loss Aversion, and Social Proof in your own words	**1 hr**
3	Dataset & ML Model Walkthrough	Explain dataset pipeline, feature engineering, and pCTR model training	**1 hr**
4	NLP Layer Deep Dive	Explain VADER, BERTScore, and NLI — what each measures and why all three are needed	**1 hr**
5	Multi-Agent System Architecture	Walk through the LangChain pipeline, agent roles, and end-to-end data flow	**1.5 hrs**
6	Results Interpretation & Evaluation	Present and discuss model evaluation metrics and output quality	**1 hr**
7	Viva Preparation — Technical Q&A	Confidently answer examiner questions on methodology and design decisions	**1.5 hrs**
8	Final Presentation Coaching	Deliver a clear, confident project presentation within the time limit	**1.5 hrs**

Paid Sessions (Optional)

For topics requiring deeper line-by-line explanation or additional mock viva practice beyond the included sessions.

Session Type	Rate (LKR)	Duration
System architecture deep dive (additional)	2,000	**1–1.5 hrs**
ML model internals & code walkthrough	2,000	**1–1.5 hrs**
NLP pipeline code walkthrough (line by line)	1,500	**1–1.5 hrs**
Evaluation metrics (F1, ROC-AUC, confusion matrix)	1,500	**1–1.5 hrs**
Mock Viva / Q&A Practice	2,500	**1–2 hrs**

Session Type	Rate (LKR)	Duration
Thesis Defense Coaching	2,500	**1–2 hrs**

4. Cost Breakdown

Deliverable	Details	Price (LKR)
System Prototype Development	Multi-agent pipeline, ML model, NLP layer, FastAPI backend, demo	**40,000**
Full Thesis Document	All 6 chapters — Introduction through Conclusion, IEEE references	**25,000**
Supporting Documentation	Architecture diagrams, evaluation report, README	**5,000**
Mentorship Sessions (× 8)	8 guided sessions covering theory, system, results, and viva prep	**15,000**
TOTAL		**LKR 85,000**

5. Payment Terms

STAGE 1 Due Now

30%

LKR 25,500

Interim Submission 01 Delivered

Chapters 1, 2 & 3 complete — thesis DOCX submitted to university

STAGE 2

40%

LKR 34,000

System Prototype Delivered

All agents + FastAPI endpoint live + Chapters 4 & 5 written (~21 Aug 2026)

STAGE 3

30%

LKR 25,500

Final Delivery

Complete thesis (Ch 6 & 7) + private GitHub repository handed over (~28 Aug 2026)

— Each stage payment is due within 48 hours of the corresponding milestone delivery
— Paid mentorship sessions are prepaid before booking
— All sessions conducted via Google Meet / Zoom

Additional Charges

Item	Rate
Urgent delivery (less than 50% of agreed timeline)	**+25% of package price**
Additional mentorship sessions	**As per paid session rates above**
Scope changes after project kickoff	**Quoted separately**

6. Terms & Conditions

1. This quotation is valid for 30 days from the date of issue.
2. Payments are structured in three stages tied to deliverable milestones — Stage 1 (30%) on Interim Submission, Stage 2 (40%) on system prototype + Ch4&5, Stage 3 (30%) on final thesis + repository. Each payment is due within 48 hours of milestone delivery.
3. Each stage's deliverables are handed over upon confirmation of the corresponding stage payment. Next-stage work begins only after the current stage payment is received.
4. Included mentorship sessions commence after Stage 3 (final delivery) payment is received.
5. Quick doubts via WhatsApp / chat are supported. Detailed explanations require a paid session.
6. All sessions are conducted online (Google Meet / Zoom). In-person sessions require prior arrangement.
7. Client is responsible for their own academic submission and integrity.
8. Source code and documents are for client's personal academic use only and shall not be resold or redistributed.
9. Any scope changes after project kickoff may result in revised pricing.
10. This quotation covers a research prototype — not a production-grade or commercially deployed system.

7. How to Proceed

1. Confirm via WhatsApp or Email to proceed
2. **Stage 1 (LKR 25,500)** — Pay within 48 hrs of receiving Interim Submission 01 (Ch1+2+3 DOCX) — *due now*
3. **Stage 2 (LKR 34,000)** — Pay within 48 hrs of receiving the working system prototype + Chapters 4 & 5 (~21 Aug 2026)
4. **Stage 3 (LKR 25,500)** — Pay within 48 hrs of receiving final thesis (Ch6+7) + private GitHub repository (~28 Aug 2026)
5. Mentorship sessions (8 included) scheduled at your convenience after Stage 3

8. Contact Information

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Thank you for your interest. Looking forward to working with you!